Anshul Kumaria

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GitHub

SKILLS

Programming Languages: Python, Java, C++, C, Swift, JavaScript, HTML5, CSS3, Xcode
Frameworks & Libraries: Django, Flask, TensorFlow, Keras, Scikit-learn, Streamlit
Database Technologies: MySQL
Machine Learning: Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Support Vector Machines (SVMs), Image Processing, Feature Extraction, Data Preprocessing
Tools & Platforms: Linux, Git, Jupyter Notebook, Google Colab

EDUCATION

MIT World Peace University 2023 – Present | Pune, India
Bachelor of Technology in Computer Science
CGPA: 9.1 / 10

PROFESSIONAL EXPERIENCE

Softech It Solution 08/2025 – 11/2025 | Pune, India
Full Stack Engineer - Trainee
- Contributed to a **production-grade School Management System** built on the **Python–Flask web framework**, supporting schools and institutions.

Staff & Faculty Management :
- Built **staff profile management** features, capturing qualifications, experience, and contact information.
- Implemented **attendance tracking**, **teacher scheduling**, and **class assignment** modules with role-based access and validation.

Fee & Financial Management :
- Developed **financial analytics dashboards** offering real-time insights into revenue, outstanding fees, and collection efficiency.

Student Management :
- Engineered **end-to-end admission and enrollment workflows** with robust **form validation** using **WTForms** and **Flask-WTF**.
- Built **attendance tracking**, **grade management**, and **report card generation** using dynamic, server-rendered templates.

Key Outcomes :
- Reduced **administrative workload** in attendance, fee tracking, and scheduling by **60%**.
- Improved **operational visibility** and reporting accuracy through automated financial and academic dashboards.

Tech Stack Used :
Backend: Python, Flask, Flask-SQLAlchemy, Flask-WTF, WTForms.
Database: MySQL / PostgreSQL / SQLite.
Frontend: HTML5, CSS3, Jinja2 Templates, JavaScript.

Infosys Springboard 11/2025 – 02/2026 | Pune, India

Artificial Intelligence Intern

AI TraceFinder

Printer/Scanner Source Identification System 2025

- Developed an **end-to-end machine learning system** to identify printer/scanner brands (Canon, Epson, HP) from scanned document images.
- Built a custom dataset using high-quality TIFF images and implemented a consistent preprocessing pipeline (grayscale conversion, resizing, normalization).
- Implemented a **Support Vector Machine (SVM)** baseline using texture-based features for initial benchmarking.
- Designed and trained a **Convolutional Neural Network (CNN) from scratch** to automatically learn scanner-specific artifacts such as noise patterns and texture inconsistencies.
- Saved trained models for reuse and deployed the final CNN using a **Streamlit-based GUI application** for real-time image inference.
- Demonstrated the effectiveness of deep learning for forensic image analysis, where device-specific patterns are imperceptible to the human eye.

Tech Stack Used:
Backend: Python.
Machine Learning & AI: TensorFlow, Keras, Scikit-learn.
Computer Vision & Image Processing: OpenCV, Pillow.
Frontend & Deployment: Streamlit.
Processing & Analysis: NumPy, Pandas.

Vodafone Idea Foundation - LLM Intern 09/2025 – 10/2025 | Pune, India

Internship

Performed in-depth data analytics and visualization on Netflix's global content catalog to uncover insights into content distribution, genre evolution, and country-wise representation.

Key Contributions:
1. Conducted extensive **Exploratory Data Analysis (EDA)** on Netflix's open dataset to examine the growth trends of Movies vs. TV Shows across different years.
2. Extracted and visualized **genre popularity patterns** using **bar charts, heatmaps, and correlation matrices** to highlight evolving viewer preferences.
3. Developed **country-wise visualizations** illustrating Netflix's global expansion and identifying underrepresented regions.
4. Delivered **business insights** into **content acquisition and production strategies**, emphasizing regional diversity and genre optimization.
5. Presented findings through a **data-driven presentation deck** combining visual storytelling with actionable recommendations.

Outcomes & Impact:
1. Uncovered **genre evolution patterns** influencing Netflix's audience engagement and regional strategy.
2. Produced **visual dashboards and plots** supporting decision-making for **content localization and expansion**.
3. Demonstrated proficiency in **data wrangling, visual analytics, and storytelling** with data using Python's analytical ecosystem.

Tech Stack Used:
Programming: Python.

Data Processing & Analysis: Pandas, NumPy.
Data Visualization: Matplotlib, Seaborn.

PROJECTS

SyncWave 08/2025 – Present
Real-Time Diarization & Accessibility App

- Architected a comprehensive iOS accessibility platform for users with Auditory Neuropathy Spectrum Disorder (ANSD), integrating real-time, multi-speaker captioning and collaborative networking for group sessions.
- Engineered a zero-latency audio processing pipeline leveraging CoreML, the Accelerate framework, and Apple Speech to achieve immediate speaker tracking (diarization), dynamic sentence segmentation, and on-device AI transcript refinement.
- Designed a highly-scalable, offline-first data architecture using SwiftData and Firebase (Firestore & Authentication) to securely synchronize conversation histories, custom voice profiles, and AI-generated post-conversation summaries.

Tech Stack Used:
Programming: Swift.
iOS Development: UIKit, SwiftData.
Machine Learning & AI: CoreML, FoundationModels.
Audio & Speech Processing: Speech, Accelerate.
Backend & Cloud: Firebase (Firestore, Authentication).
Development Tools: Xcode.

Sub-Atomica 02/2026 – Present
Quantum Mechanics Visualiser

- **Designed and developed** an immersive augmented reality educational app teaching quantum mechanics using Swift, SwiftUI, RealityKit, and ARKit.
- **Implemented** real-time 3D physics simulations (e.g., superposition, entanglement) with dynamic user controls, spatial plane tracking, and an interactive "Lab Notebook" for capturing AR snapshots.
- **Built** a high-fidelity, aerospace-themed interface featuring on-device machine learning (Vision framework) for simulated data analysis and custom haptics for a highly tactile user experience.

Tech Stack Used:
Programming: Swift.
UI Development: SwiftUI.
Augmented Reality & 3D: RealityKit, ARKit.
Machine Learning & Vision: Vision.
Media & Audio: AVFoundation.
Development Tools: Xcode.

A.N.S.H.U.L. 04/2022 – Present
A Nice Sophisticated Helpful Unique Listener
A voice-activated assistant that leverages speech recognition and task automation to streamline various activities.

- **Messaging**: Utilizes the pywhatkit library to schedule and send WhatsApp messages.
- **Web Searching**: Employs a web browser to open browser tabs for searches, providing immediate online data retrieval.
- **Music Playback**: Directs users to an online playlist to fulfill entertainment-related commands.

- **Date and Time Queries**: Functions to retrieve and verbally communicate the current day and time are implemented, enhancing the assistant's usability as a real-time utility tool.

Tech Stack Used:
Programming: Python.
Automation & Scripting: PyWhatKit.
Speech & Voice Processing: SpeechRecognition, Text-to-Speech.
Web Integration: Web Browser APIs.

Medi-Master 01/2025 – 04/2025
Medication Recommendation Engine
Built a full-stack medical record management system that allows users to **enter symptoms to receive mapped prescriptions** and **view historical medical records**, with **secure multi-role access** and **optimized SQL procedures** ensuring data consistency.

Tech Stack Used*:*
Backend: Python, Flask.
Frontend: HTML, CSS, JavaScript.
Database: MySQL.

Credit Card Fraud Detection 10/2025 – 10/2025
ML-based Analysis
- Built a machine learning model to detect fraudulent credit card transactions using the Kaggle dataset.
- Implemented extensive preprocessing to handle imbalanced data and tested multiple classifiers (Logistic Regression, Random Forest, XGBoost, etc.).
- Achieved over **99% accuracy** and **high recall (>90%)** using Random Forest.
- Visualized data distributions, correlation heatmaps, and confusion matrices for interpretability.

Tech Stack Used:
Programming: Python.
Data Processing & Analysis: Pandas, NumPy.
Machine Learning: Scikit-learn, XGBoost.
Data Visualization: Matplotlib, Seaborn.